Stress Test of Candidate Analyses for the Biomarker-by-Drug Interaction in the Hybrid Design: Exploratory Results

2026-10-08 09:49 PDT

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Purpose. This document reports the results of the stress test set out in the working protocol `docs/46`. Fourteen analyses of the biomarker-by-drug interaction $\beta_{bm:D}$ in the Hybrid design were fitted to the same 28,000 simulated trials in 41 cells, and judged for size, robustness to a prognostic biomarker, sensitivity and power. The study is exploratory: its criteria and its set of analyses were revised as results came in (`docs/46`, Amendments 1 and 2), so the selected analysis is a recommendation to be confirmed in an independent study. The reasoning behind several revisions is in `docs/48`; the recommendation for practice is in `docs/49`.

Rerun note (2026-10-11). Sections 1 to 11 are the stored run (recursive carryover, fixed allocation; 2026-10-07), and are left as written. The main run (eleven interaction analyses) has been repeated under the refined process, with anchored carryover and randomized path allocation; see the section "Rerun under the refined process (2026-10-11)" before the references. In brief: size and the prognostic flags are unchanged within Monte Carlo error; DS and EA now narrowly fail the size screen; among the eleven analyses F1 remains the unflagged survivor with the smallest maximin shortfall (0.240, stored 0.242); and at the AR(1) reference cell F1 has 11% less size-adjusted power than S (0.448 against 0.504), not the 29% of the stored run.

Decision note (2026-10-15). G1, G2 and F1A have since been rerun on the trials of the main rerun (jobs 21 and 22), and 1,500 fresh trials per cell were added for the eight surviving analyses (job 37, post hoc). F1 fails nothing; F1A fails the size screen. On the 28,000 rerun trials G2 led F1 by 0.026 in maximin shortfall, beyond the tie margin of 0.02; on the fresh trials alone and on all trials pooled, F1, G1 and G2 lie within the margin (pooled 0.216, 0.221 and 0.221), and the tie-break selects **F1**. F1 is retained as the recommended analysis under the refined process; G1 and G2 are post hoc alternatives that tie with it, and the earlier G2 lead is attributed to Monte Carlo error. See the subsections "Fourteen-analysis decision" and "Fresh trials for the F1, G1, G2 choice" of the rerun section.

Contents

Notation index and glossary 4
1. Summary 6
2. Design in brief 6
3. Size 7
4. Robustness to a prognostic biomarker 8
5. Sensitivity 9
6. Power 10
7. Selection 11
8. Choosing the half-life by AIC 13
9. Failed fits 13
10. Limitations 13
11. Reproducibility 14
Rerun under the refined process (2026-10-11) 14
What changed in the data-generating process 14
Size 15
Survivors 17
Maximin shortfall 18
Power 18
Fourteen-analysis decision (jobs 21 and 22; added 2026-10-15) . . 21
Fresh trials for the F1, G1, G2 choice (job 37; added 2026-10-15) 22
Decision (2026-10-15) 24
12. References 24

List of Tables

1 Notation index and glossary of symbols used in this document . . 4
2 The fourteen analyses compared in the stress test 5
3 Null rejection rates by analysis: pooled over the 11 non-prognostic null cells, largest single cell, and the four prognostic null cells 7

4 Size-adjusted power curves under AR(1): monotonicity in both constructs, power at plus and minus 0.30, steepness between 0.15 and 0.30, and the smallest coupling reaching power 0.80 9

5 Size-adjusted power (unadjusted in parentheses) in the two reference alternative cells, $N = 70$ 10

6 Size-adjusted power at $c_{bm} = 0.30$ in the stress cells: the reference analysis S, the most powerful analysis C1, and the five robust analyses 11

7 Maximin shortfall in size-adjusted power among the analyses that pass the size and sensitivity screens, over the non-prognostic alternative cells and over all alternative cells 12

8 Share of fits choosing each analysis half-life, over all cells 13

9 Rerun under the refined process: null rejection rates of the eleven main-run analyses over the 11 non-prognostic null cells (pooled, 2-MCSE band 0.0458 to 0.0542; largest single cell, cap 0.075), stored run and rerun 15

10 Rerun under the refined process: null rejection rates in the four prognostic null cells (rerun, stored run in parentheses; cap 0.075) and the prognostic flag, which is the same in both runs 16

11 Rerun under the refined process: analyses passing the screens, stored run and rerun 17

12 Rerun under the refined process: maximin shortfall in size-adjusted power among the survivors of the size and sensitivity screens (eleven analyses; the best analysis in each cell is taken over all survivors, flagged ones included) 18

13 Rerun under the refined process: size-adjusted power (unadjusted in parentheses) in the two reference alternative cells, $N = 70$, rerun and stored run 19

14 Rerun under the refined process: size-adjusted power at $c_{bm} = 0.30$ in the stress cells for S, C1 and the two unflagged survivors F1 and ES (stored run in parentheses) 19

15 Rerun under the refined process, fourteen analyses: size, prognostic flag and maximin shortfall of the robust analyses (stored run in parentheses; the shortfall is measured against the best survivor in each cell, flagged survivors included) 21

16 Job 37 and job 37 pooled with jobs 15 and 21: size over the 11 non-prognostic null cells, prognostic flag and maximin shortfall in size-adjusted power (the shortfall is measured against the best survivor in each cell; the bootstrap SE is reported for the three analyses in contention) 23

Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here or in `docs/46`.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t (code `Sx`); $t = 0$ is baseline	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
B_i^c	biomarker centered at the trial's baseline sample mean, $B_i - \bar{B}$, not scaled (code `bmc`, the form the analyses use; see the note below the table)	canonical
$D_{bc,it}$	exposure-decayed drug indicator, 0 before any exposure (code `Dbc`)	canonical
η_t	expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `De`)	canonical
$t_{1/2}$, $t_{1/2}^a$	true carryover half-life; half-life assumed by the analysis	canonical; local
c_{bm}	biomarker-BR coupling, the predictive effect	canonical
$c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$	biomarker correlations with the natural course, the placebo response and baseline severity (prognostic couplings)	canonical
β_{bm}^{TV}	mean moderation of the natural course by b_i (growing prognostic pathway)	canonical
$\beta_{bm:D}$	biomarker-by-drug interaction, the estimand	canonical
ρ	AR(1) serial correlation per week	canonical
N	total sample size across paths	canonical
α	nominal level, 0.05	canonical
π, $\hat{\pi}$	power; an estimated rejection rate (size or power)	canonical
n_{reps}	Monte Carlo replicates per cell	canonical

(Corrected 2026-10-08: `bmc` was described here as the standardized biomarker. In `11-strawman-stress-test.R` it is $B_i - \bar{B}$, centered at the trial's baseline sample mean and not divided by an SD. Test statistics and p-values are invariant to rescaling the biomarker, so no reported rate changes; the fitted $\beta_{bm:D}$ is per unit of the raw biomarker. The terms `bmc:De` are therefore written $B_i^c \eta_t = (B_i - \bar{B})\eta_t$ below (renamed from $b_i \eta_t$ in the notation pass of 2026-10-09). See `analysis/report/compendium-mathematics.tex`, Section "The stress-test construct and its fourteen analyses".)

The fourteen analyses (`docs/46`, Section 4 and Amendment 2):

Table 2: The fourteen analyses compared in the stress test

ID	Analysis
S	random intercept plus continuous-time AR(1) residuals (`corCAR1`), CR2 standard errors with Satterthwaite degrees of freedom, baseline as a response row, linear time, $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week
DS	S with $t^a_{1/2}$ chosen by AIC over {0, 0.25, 0.5, 1, 2}
C1, C2	S with residual variances by phase (baseline, open label, blinded); CR2 (C1) or model-based (C2) inference
F1	S plus the expectancy terms η_t and $B^c_i \eta_t$ (`De + bmc:De`)
F1A	F1 with $t^a_{1/2}$ chosen by AIC
G1, G2	F1 with residual variances by phase; CR2 (G1) or model-based (G2) inference
F2	S plus `bmc:t`
ES, EA	S and A with the biomarker's baseline association freed (`+ bmc:bl0`)
A	unstructured covariance with Kenward-Roger inference, otherwise as S
DA	A with $t^a_{1/2}$ chosen by AIC
R	the published reference arm: compound-symmetry covariance (the `cs` structure of `mmrm`, by visit within participant, equivalent to a random intercept with independent errors), model-based standard errors with Satterthwaite degrees of freedom, binary drug indicator (wording corrected 2026-10-08; `docs/46` described it as a random intercept with model-based inference)

Glossary.

- **Null and alternative cells.** A null cell has $c_{bm} = 0$ (1,000 replicates); its alternative twin has $c_{bm} = 0.30$, or 0.25 under compound symmetry (500 replicates).
- **Prognostic cell.** A cell in which the biomarker is associated with the outcome regardless of treatment: through the natural course (constant, $c_{bm,TV} = 0.3$, or growing, $\beta^{TV}_{bm} = 0.3$), the placebo response ($c_{bm,PB} = 0.3$) or baseline severity ($c_{bm,BL} = 0.3$).
- **Size-adjusted power.** Power when each analysis rejects at the 5% quantile of its own p-values in the matching null cell.
- **Shortfall.** In an alternative cell, the gap between an analysis's size-adjusted power and that of the best nominal analysis; the *maximin shortfall* of an analysis is its largest shortfall over the cells considered. (Clarified 2026-10-08: the term is kept because the decision outputs use it; selecting the analysis with the smallest largest shortfall is a minimax-regret rule. The best analysis in each cell is taken over all survivors of the size and sensitivity screens, including the flagged C1 and C2, so shortfalls are measured against the phase-variance models. See

`analysis/report/compendium-mathematics.tex`, Section "Size, power screens and the selection criteria of the stress test".)

- **MCSE.** Monte Carlo standard error, $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for a rejection rate $\hat{\pi}$ over n_{reps} replicates.

1. Summary

All results are from stored simulation outputs (verified; Section 11).

1. **Size.** Ten of the fourteen analyses hold their size over the 11 non-prognostic null cells (pooled 0.047 to 0.053, within the 2-MCSE band 0.046 to 0.054, and no cell above 0.075). Four fail: the unstructured model A (0.057 pooled, 0.087 at $N = 35$), its AIC version DA (0.065), the published reference arm R (0.069), and F1A (0.057, and 0.073 when $t_{1/2} = 1$ week).
2. **A prognostic biomarker is the decisive stress.** Every analysis that shares the biomarker's main effect between the baseline row and the later visits rejects too often in at least one prognostic null cell: S 0.100, C1 0.382, A 0.416. Five analyses stay within the cap in all four prognostic null cells: ES, EA, F1, G1 and G2.
3. **Sensitivity.** Every analysis has monotone size-adjusted power curves in both constructs. The symmetry screen was dropped, because power is symmetric in the sign of c_{bm} by construction.
4. **Power.** Robustness costs power. At $c_{bm} = 0.30$ under AR(1), size-adjusted power is 0.514 for S and 0.364 for F1, 29% less; under compound symmetry at $c_{bm} = 0.25$ the difference is small (0.554 against 0.522). F1 has significantly lower power than DS in 15 of the 22 non-prognostic alternative cells (McNemar test at 0.05) and higher power in none.
5. **Selection.** Among the five robust analyses the maximin shortfall over the non-prognostic alternative cells is G2 0.232, F1 0.242, G1 0.252, ES 0.278 and EA 0.318. G2 and F1 lie within the tie margin of 0.02, neither reaches power 0.80 on the grid, and the rule's last tie-break (fewer covariance parameters) selects **F1**. G1 and G2 are effectively equivalent alternatives.
6. **A second reading.** Over all 26 alternative cells, prognostic cells included, G1 (0.308) and G2 (0.312) lead F1 (0.358) by about 0.05. The protocol does not say which set of cells governs the selection, so both readings are reported (Section 7).

2. Design in brief

The design is set out in `docs/46`; this section records what was run.

- **Design.** Hybrid, visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 plus baseline, four paths with the published allocation.

- **Data-generating process.** Configuration C of `docs/36`: separable AR(1) correlation at $\rho = 0.7$, graded coupling of the biomarker to the drug response, the published recursive carryover on the drug-response mean, the published Gompertz means and SDs, the fixed published allocation, and a stop on any matrix that is not positive definite; base cell $N = 70$, $t_{1/2} = 0.5$ week, no dropout, exponential decay. (Corrected 2026-10-08: this bullet said "anchored carryover". `11-strawman-stress-test.R` decays the previous visit's mean with the cumulative time since discontinuation, the published recursion (inspected). The construct is therefore the covariance construct of the refined process of paper 01, not the refined process in full.)
- **Cells.** 41: two power curves (AR(1), c_{bm} from -0.30 to 0.45; compound symmetry, -0.20 to 0.25) and stress cells varying one factor each (measurement error, $N = 35$ and 150, $t_{1/2} = 0$ and 1 week, Weibull decay with shape 0.5 and 2, 20% MCAR and MAR dropout, and the four prognostic pathways). There are 15 null and 26 alternative cells and 28,000 simulated trials.
- **Analyses.** All fourteen are fitted to the same trials, so every comparison is paired.
- **How the protocol changed** (`docs/46`, Amendments 1 and 2):
    1. the reference analysis became DS, and the symmetry screen was dropped;
    2. the two baseline-severity cells (40 and 41) were added and judged with the other prognostic cells;
    3. G1, G2 and F1A were added and judged by the same rule;
    4. failed fits are excluded from rejection rates, rather than counted as non-rejections as first written; the two conventions differ by at most 0.0025 in any rate.

3. Size

Table 3: Null rejection rates by analysis: pooled over the 11 non-prognostic null cells, largest single cell, and the four prognostic null cells

Analysis	Pooled	Largest cell	Nominal	TV, constant	TV, growing	Placebo	Baseline	Prognostic flag
S	0.048	0.059 (MCAR dropout)	yes	0.099	0.049	0.100	0.050	yes
DS	0.053	0.068 (MCAR dropout)	yes	0.115	0.058	0.112	0.055	yes
C1	0.053	0.069 ($N = 35$)	yes	0.382	0.173	0.205	0.054	yes
C2	0.052	0.066 ($N = 35$)	yes	0.393	0.172	0.193	0.050	yes

Analysis	Pooled	Largest cell	Nominal	TV, constant	TV, growing	Placebo	Baseline	Prognostic flag
F1	0.051	0.062 (AR(1) curve)	yes	0.047	0.046	0.040	0.040	no
F1A	0.057	0.073 ($t_{1/2} = 1$)	no	0.059	0.052	0.047	0.045	no
G1	0.049	0.060 (AR(1) curve)	yes	0.055	0.044	0.042	0.042	no
G2	0.047	0.066 (AR(1) curve)	yes	0.062	0.043	0.045	0.046	no
F2	0.049	0.059 (MCAR dropout)	yes	0.090	0.048	0.098	0.048	yes
ES	0.048	0.059 (AR(1) curve)	yes	0.049	0.063	0.049	0.050	no
EA	0.052	0.070 ($N = 35$)	yes	0.049	0.057	0.050	0.066	no
A	0.057	0.087 ($N = 35$)	no	0.416	0.195	0.222	0.051	yes
DA	0.065	0.103 ($N = 35$)	no	0.509	0.245	0.256	0.064	yes
R	0.069	0.081 ($N = 35$)	no	0.107	0.053	0.159	0.075	yes

The pooled rate is over 11,000 replicates, so its 2-MCSE band is 0.046 to 0.054; a single cell has 1,000 replicates (MCSE 0.0069), and the criterion caps each cell at 0.075. An analysis is flagged when any prognostic null cell exceeds 0.075.

Reading.

- **Inference matters most at small N.** The unstructured model with Kenward-Roger inference [5] rejects 0.087 at $N = 35$, and with an AIC-chosen half-life 0.103. The random-intercept models with CR2 standard errors [3, 4] stay at or below 0.069 in every cell, except F1A (0.073), whose half-life is chosen by AIC.
- **Choosing the half-life by AIC inflates size slightly.** DS (0.053) passes; F1A (0.057) and DA (0.065) do not. The selection step adds variability that the standard errors do not reflect.
- **The published reference arm is liberal** throughout (0.069 pooled), as earlier documents found (`docs/45`).
- **Baseline severity alone does little harm.** A prognostic association with baseline severity enters every visit equally, so every analysis stays within the cap in that cell (0.040 to 0.075; R reaches the cap exactly, and EA reaches 0.066).

4. Robustness to a prognostic biomarker

Analyses that keep baseline as a response row and use one biomarker main effect for it and for the later visits attribute part of a prognostic association to the

interaction (derived, `docs/48`, Section 2). The size table shows this directly: S, DS, C1, C2, F2, A, DA and R are flagged, with null rejection rates up to 0.509 when the natural course is associated with the biomarker. F2's `bmc:t` term remedies the growing natural-course pathway (0.048) but not the constant one (0.090) or the placebo pathway (0.098).

Two families stay within the cap in every prognostic cell:

- **the expectancy-term models** F1, G1 and G2, which add η_t and $B_i^c \eta_t$ and so let the biomarker's association differ between baseline, open-label and blinded visits;
- **the free-baseline models** ES and EA, which give the baseline row its own biomarker slope.

Staying within the cap is an empirical result, not exact invariance. In the Hybrid design F1 leaves a small residual bias under a constant natural-course pathway, because the blinded on-drug and off-drug visits carry unequal weight, and ES leaks a little under the placebo pathway through the open-label visits (derived; `docs/49`, Section 3, and `docs/48`, Section 2).

The protocol's rule recommends a flagged analysis only together with a remedy (ES, EA, F1 or F2) that is nominal in the failing cell. The remedies that pass every prognostic cell are themselves candidates, so the selection is made among the unflagged analyses.

5. Sensitivity

Table 4: Size-adjusted power curves under AR(1): monotonicity in both constructs, power at plus and minus 0.30, steepness between 0.15 and 0.30, and the smallest coupling reaching power 0.80

Analysis	Monotone (AR(1) / CS)	Power at $c_{bm} = -0.30$ / $+0.30$	Steepness	Detectable coupling
S	yes / yes	0.450 / 0.514	2.32	0.446
DS	yes / yes	0.422 / 0.486	2.21	not reached
C1	yes / yes	0.524 / 0.578	2.49	0.403
C2	yes / yes	0.554 / 0.606	2.65	0.403
F1	yes / yes	0.352 / 0.364	1.80	not reached
F1A	yes / yes	0.326 / 0.338	1.67	not reached
G1	yes / yes	0.340 / 0.354	1.72	not reached
G2	yes / yes	0.360 / 0.380	1.75	not reached
F2	yes / yes	0.446 / 0.516	2.36	not reached
ES	yes / yes	0.336 / 0.364	1.71	not reached

Analysis	Monotone (AR(1) / CS)	Power at $c_{bm} = -0.30$ / $+0.30$	Steepness	Detectable coupling
EA	yes / yes	0.300 / 0.306	1.48	not reached
A	yes / yes	0.510 / 0.534	2.48	0.430
DA	yes / yes	0.472 / 0.498	2.25	0.434
R	yes / yes	0.394 / 0.398	1.76	not reached

Reading.

- **Every curve is monotone**, so the sensitivity screen removes no analysis.
- **The symmetry screen was defective.** The data-generating process treats $+c_{bm}$ and $-c_{bm}$ alike, so power is symmetric in theory; the observed differences between -0.30 and $+0.30$ (at most 0.070, F2, against an MCSE of about 0.03 for a difference of two independent cells of 500) are consistent with Monte Carlo error, and the screen would have eliminated analyses for noise. It was dropped in Amendment 1 (derived, `docs/48`, Section 4).
- **The robust analyses are flatter.** Their steepness is 1.5 to 1.8, against 2.2 to 2.7 for the flagged analyses other than R (1.76), and none reaches power 0.80 by $c_{bm} = 0.45$ at $N = 70$. Steepness and detectable coupling therefore cannot break ties among them.

6. Power

Table 5: Size-adjusted power (unadjusted in parentheses) in the two reference alternative cells, $N = 70$

Analysis	AR(1), $c_{bm} = 0.30$	Compound symmetry, $c_{bm} = 0.25$
S	0.514 (0.508)	0.554 (0.554)
DS	0.486 (0.516)	0.480 (0.476)
C1	0.578 (0.602)	0.660 (0.658)
C2	0.606 (0.606)	0.656 (0.630)
F1	0.364 (0.410)	0.522 (0.544)
F1A	0.338 (0.436)	0.446 (0.508)
G1	0.354 (0.408)	0.546 (0.552)
G2	0.380 (0.426)	0.528 (0.504)
F2	0.516 (0.496)	0.552 (0.552)
ES	0.364 (0.412)	0.662 (0.634)
EA	0.306 (0.392)	0.602 (0.596)
A	0.534 (0.566)	0.618 (0.618)
DA	0.498 (0.592)	0.572 (0.596)
R	0.398 (0.476)	0.596 (0.594)

Analysis	AR(1), $c_{bm} = 0.30$	Compound symmetry, $c_{bm} = 0.25$

Table 6: Size-adjusted power at $c_{bm} = 0.30$ in the stress cells: the reference analysis S, the most powerful analysis C1, and the five robust analyses

Cell	S	C1	F1	G1	G2	ES	EA
$N = 150$	0.828	0.918	0.748	0.752	0.754	0.704	0.698
$N = 35$	0.222	0.272	0.192	0.214	0.218	0.206	0.196
$t_{1/2} = 0$	0.594	0.666	0.496	0.510	0.532	0.504	0.490
$t_{1/2} = 1$ week	0.388	0.502	0.272	0.282	0.306	0.288	0.294
Weibull decay, shape 0.5	0.404	0.514	0.308	0.296	0.378	0.270	0.230
Weibull decay, shape 2	0.588	0.682	0.504	0.526	0.526	0.458	0.464
Measurement error	0.400	0.444	0.318	0.320	0.330	0.336	0.324
20% MCAR dropout	0.426	0.542	0.378	0.394	0.406	0.360	0.346
20% MAR dropout	0.418	0.518	0.382	0.404	0.378	0.356	0.322
Prognostic, TV constant	0.606	0.774	0.482	0.520	0.524	0.370	0.378
Prognostic, TV growing	0.516	0.672	0.378	0.484	0.490	0.192	0.208
Prognostic, placebo	0.652	0.736	0.446	0.428	0.424	0.498	0.438
Prognostic, baseline	0.444	0.552	0.396	0.394	0.406	0.336	0.290

Reading.

- **The phase-variance models are the most powerful** (C1 has the highest power in every stress cell shown), but they fail the prognostic cells badly, so their power is not available to a trial whose biomarker may be prognostic.
- **The cost of robustness depends on the correlation structure.** Under AR(1) the robust analyses have 26% (G2) to 40% (EA) less size-adjusted power than S at $c_{bm} = 0.30$; under compound symmetry ES even exceeds S (0.662 against 0.554), and F1 is close (0.522).
- **The power reported for flagged analyses in prognostic cells is size-adjusted**, so it is not the power a trial would see: unadjusted, those analyses reject far too often under the null.
- **The free-baseline models lose most when the natural course grows with the biomarker** (0.192 and 0.208), which is why their all-cells shortfall is large (Section 7).

7. Selection

The rule (`docs/46`, Section 7, as revised by Amendment 1) is applied to the analyses that pass the size screen and the sensitivity screen and are not flagged in a prognostic cell: ES, EA, F1, G1 and G2.

Table 7: Maximin shortfall in size-adjusted power among the analyses that pass the size and sensitivity screens, over the non-prognostic alternative cells and over all alternative cells

Analysis	Largest shortfall, non-prognostic cells	Worst cell	Largest shortfall, all cells	Worst cell
C1 (flagged)	0.034	Weibull 0.5	0.064	TV growing
C2 (flagged)	0.040	Weibull 2	0.040	Weibull 2
S (flagged)	0.144	Weibull 0.5	0.220	TV growing
F2 (flagged)	0.160	Weibull 0.5	0.244	TV growing
DS (flagged)	0.182	CS curve, 0.25	0.214	TV growing
G2	0.232	AR(1) curve, 0.375	0.312	placebo
F1	0.242	AR(1) curve, 0.30	0.358	TV growing
G1	0.252	Weibull 0.5	0.308	placebo
ES	0.278	Weibull 0.5	0.544	TV growing
EA	0.318	Weibull 0.5	0.528	TV growing

Outcome.

1. **Smallest shortfall.** Over the non-prognostic alternative cells, G2 (0.232) and F1 (0.242) are the two best robust analyses.
2. **Tie.** They differ by 0.010, inside the rule's tie margin of 0.02.
3. **First tie-break.** The smaller detectable coupling in the base construct is undefined: neither reaches power 0.80 on the grid.
4. **Last tie-break.** The analysis with fewer covariance parameters, **F1**, is selected. F1 is also the only one of the two with CR2 inference; G2 relies on model-based standard errors.
5. **Strawman rule.** The reference analysis DS is flagged in the prognostic cells, so it cannot be retained; F1 replaces it. The paired comparison makes the price explicit: F1 has significantly lower size-adjusted power than DS in 15 of the 22 non-prognostic alternative cells, and higher power in none.

The all-cells reading. If the prognostic alternative cells are included in the maximin set, the order among the robust analyses is G1 0.308, G2 0.312, F1 0.358, EA 0.528, ES 0.544, and the phase-variance versions lead F1 by about 0.05. The working protocol does not say whether those cells count. The non-prognostic reading is reported as the outcome because the prognostic cells are

judged by their own screen. Under the all-cells reading G1 and G2 would tie (0.004 apart) ahead of F1, and the tie-breaks could not separate them, since they share one model and differ only in inference. The difference from F1 arises mainly when the natural course is associated with the biomarker (G1 0.484 against F1 0.378 in that cell).

8. Choosing the half-life by AIC

DS, DA and F1A choose $t^a_{1/2}$ by AIC from {0, 0.25, 0.5, 1, 2} weeks (0 meaning a binary drug indicator), with maximum likelihood for the choice and a REML refit.

Table 8: Share of fits choosing each analysis half-life, over all cells

Analysis	0	0.25	0.5	1	2
DS	0.170	0.144	0.299	0.208	0.179
DA	0.188	0.144	0.294	0.181	0.193
F1A	0.236	0.163	0.292	0.190	0.120

The choice is spread across the grid even though the true half-life is 0.5 week in most cells, and the selection step costs size: F1A fails the size criterion where F1 passes, and DS loses about 0.03 of size-adjusted power against S at the AR(1) reference cell (0.486 against 0.514). A fixed $t^a_{1/2} = 0.5$ week is therefore preferred with F1.

9. Failed fits

A fit that returned an error was recorded as missing. Failures number 0 to 12 per analysis over 28,000 trials (G1 and G2 12; S, C1 and C2 6; F1, F1A and ES 3; DS and F2 2; A, DA, EA and R 0), and at most 0.6% in any cell (G1 and G2), below the 2% at which an analysis would be flagged.

10. Limitations

- **Exploratory.** The criteria, the reference analysis and the set of analyses were revised after results were seen. The selection is therefore a recommendation from an exploratory comparison, and should be confirmed in an independent simulation study before it is relied on.
- **One design and one effect size.** Only the Hybrid design was studied, with the alternative cells at $c_{bm} = 0.30$ (0.25 under compound symmetry) and prognostic couplings of 0.3.

- **The set of cells governs the maximin.** The outcome depends on whether the prognostic alternative cells count (Section 7).
- **Monte Carlo error.** Alternative cells have 500 replicates (MCSE of power at most 0.022), so differences in shortfall below about 0.03 between two analyses are within noise for a single cell.
- **The prognostic mechanisms exist only in the stress-test driver,** not yet in the package (`docs/44`).

11. Reproducibility

All numbers in this document were read from stored outputs on 2026-10-08:

- `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` produced the 28,000 trials and the fits (main run, `--extra` for G1 and G2, `--extra2` for F1A); replicate and summary files are in `analysis/data/quick-sim/carryover-closed-form/strawman-stress/` (`replicates-full*.rds`, `summary-full*.csv`).
- `12-strawman-stress-decision.R --strawman DS --no-symmetry --with-extras` applies the rule to all fourteen analyses and writes `decision-screen-ds-nosym-all14.csv`, `decision-sensitivity-ds-nosym-all14.csv`, `decision-power-ds-nosym-all14.csv` and `decision-maximin-ds-nosym-all14.csv`.
- Seeds are set per chunk as $20261007 + 1000 \times \text{cell} + \text{chunk}$, so results do not depend on the number of cores.

Rerun under the refined process (2026-10-11)

Sections 1 to 11 above report the stored run (recursive carryover, fixed allocation; 2026-10-07). This section reports the main run of the stress test repeated under the refined process. It covers the eleven interaction analyses of the main run (S, DS, C1, C2, F1, F2, ES, EA, A, DA and R). The three analyses of the `--extra` and `--extra2` runs (G1, G2 and F1A) were rerun on the same trials afterwards (jobs 21 and 22), and the fourteen-analysis decision of Section 7 is repeated in the subsection "Fourteen-analysis decision" below. The subsection "Fresh trials for the F1, G1, G2 choice" reports a further, post hoc run that was made because that decision was within Monte Carlo error of a tie. (Updated 2026-10-15; the first paragraphs of this section were written on 2026-10-11, before jobs 21, 22 and 37.)

What changed in the data-generating process

Two features of the stored run departed from the refined process (`analysis/report/NOTATION.md`, "Named data-generating processes"; Section 2 of this document). Both were changed in `11-strawman-stress-test.R` (`RERUN-anchored.md`, Section 1):

1. **Anchored carryover.** Off drug, the mean drug response is $\mu_{\text{last}}\, 2^{-t_{sd}/t_{1/2}}$, with μ_{last} the mean at the last on-drug visit, weighted in the Weibull cells by the cell's decay weight. The published recursion decayed the previous visit's already-decayed mean by the cumulative time since discontinuation, so it counted elapsed time twice from the second off-drug visit of a run onward (at $t_{1/2} = 0.5$ week the recursive mean at the second off-drug visit is a quarter of the anchored one). The means of the $t_{1/2} = 0$ cell are unchanged.
2. **Randomized path allocation.** In every trial each participant is assigned to one of the four Hybrid paths uniformly at random, redrawn when a path receives fewer than 2 participants, instead of the fixed published allocation.

The 41 cells, the 28,000 trials (1,000 per null cell, 500 per alternative cell), the eleven analyses and the per-chunk seeds are those of the stored run. The trials are nevertheless not the same: the means differ, and the allocation draws consume random numbers. The stored and rerun rates are therefore independent estimates, and a difference is judged by $z = (\hat{\pi}_{\text{new}} - \hat{\pi}_{\text{old}})/\sqrt{\text{MCSE}_{\text{old}}^2 + \text{MCSE}_{\text{new}}^2}$. Within the rerun, all eleven analyses are fitted to the same trials, so comparisons between analyses remain paired.

The run is simulation-server job 15 at commit `e077526` (`replicates-full-anchored.rds`, `summary-full-anchored.csv`). The decision rule was applied with `12-strawman-stress-decision.R --strawman DS --no-symmetry --tag -anchored` (Amendment 1 reading) and with `--tag -anchored` alone (the reading as first specified), writing the screen, sensitivity, power and maximin files `decision-*-ds-nosym-anchored.csv` and `decision-*-anchored.csv`. The stored-run figures below were recomputed from `replicates-full.rds` with the same script, for the same eleven analyses and all 41 cells (verified: the recomputation with `--with-extras` reproduces the four stored `-all14` files exactly). As in the stored decision files, a failed fit counts as a non-rejection. There were 26 failed fits in 308,000 (stored run 28; 0 to 5 per analysis).

Size

Table 9: Rerun under the refined process: null rejection rates of the eleven main-run analyses over the 11 non-prognostic null cells (pooled, 2-MCSE band 0.0458 to 0.0542; largest single cell, cap 0.075), stored run and rerun

Analysis	Pooled, stored	Pooled, rerun	Largest cell, stored	Largest cell, rerun	Nominal, stored / rerun
S	0.0483	0.0516	0.059 (MCAR dropout)	0.061 ($N = 35$)	yes / yes

Analysis	Pooled, stored	Pooled, rerun	Largest cell, stored	Largest cell, rerun	Nominal, stored / rerun
DS	0.0528	0.0569	0.068 (MCAR dropout)	0.067 (Weibull 0.5)	yes / **no**
C1	0.0531	0.0529	0.069 ($N = 35$)	0.065 (meas. error)	yes / yes
C2	0.0519	0.0492	0.066 ($N = 35$)	0.055 (meas. error)	yes / yes
F1	0.0511	0.0514	0.062 (AR(1) curve)	0.064 ($t_{1/2} = 0$)	yes / yes
F2	0.0486	0.0519	0.059 (MCAR dropout)	0.060 ($N = 35$)	yes / yes
ES	0.0478	0.0490	0.059 (AR(1) curve)	0.060 (Weibull 0.5)	yes / yes
EA	0.0517	0.0575	0.070 ($N = 35$)	0.076 ($N = 35$)	yes / **no**
A	0.0573	0.0571	0.087 ($N = 35$)	0.084 ($N = 35$)	no / no
DA	0.0653	0.0656	0.103 ($N = 35$)	0.093 ($N = 35$)	no / no
R	0.0687	0.0686	0.081 ($N = 35$)	0.084 ($N = 35$)	no / no

Table 10: Rerun under the refined process: null rejection rates in the four prognostic null cells (rerun, stored run in parentheses; cap 0.075) and the prognostic flag, which is the same in both runs

Analysis	TV, constant	TV, growing	Placebo	Baseline	Prognostic flag
S	0.099 (0.099)	0.055 (0.049)	0.114 (0.100)	0.056 (0.050)	yes
DS	0.127 (0.115)	0.068 (0.058)	0.130 (0.112)	0.057 (0.055)	yes
C1	0.367 (0.382)	0.196 (0.173)	0.191 (0.205)	0.056 (0.054)	yes
C2	0.382 (0.393)	0.190 (0.172)	0.178 (0.193)	0.051 (0.050)	yes
F1	0.050 (0.047)	0.050 (0.046)	0.048 (0.040)	0.052 (0.040)	no
F2	0.094 (0.090)	0.052 (0.048)	0.112 (0.098)	0.054 (0.048)	yes
ES	0.045 (0.049)	0.073 (0.063)	0.063 (0.049)	0.057 (0.050)	no
EA	0.049 (0.049)	0.057 (0.057)	0.063 (0.050)	0.061 (0.066)	no
A	0.411 (0.416)	0.210 (0.195)	0.193 (0.222)	0.063 (0.051)	yes

Analysis	TV, constant	TV, growing	Placebo	Baseline	Prognostic flag
DA	0.515 (0.509)	0.276 (0.245)	0.229 (0.256)	0.065 (0.064)	yes
R	0.119 (0.107)	0.077 (0.053)	0.167 (0.159)	0.069 (0.075)	yes

Reading.

- **Size is unchanged within Monte Carlo error.** Over the 165 null rates (15 null cells by 11 analyses), 7 differ from the stored run by more than 2 in $|z|$, about the 8 expected by chance; the largest is EA at $t_{1/2} = 0$ (0.040 stored, 0.069 rerun, $z = 2.9$).
- **Two analyses cross a boundary.** DS (pooled 0.0569, outside the band) and EA (pooled 0.0575, and 0.076 at $N = 35$ against the cap of 0.075) now fail the size screen. Neither change is large against Monte Carlo error ($z = 1.4$ and 2.0 for the pooled rates); both were close to the boundary in the stored run. A, DA and R fail as before.
- **The prognostic flags are unchanged** for every analysis. F1 stays at or below 0.052 in all four prognostic null cells. ES stays within the cap with a narrow margin in the growing natural-course cell (0.073, MCSE 0.008).

Survivors

Every size-adjusted power curve is again monotone in both constructs, so the sensitivity screen (Amendment 1, no symmetry) removes no analysis.

Table 11: Rerun under the refined process: analyses passing the screens, stored run and rerun

Reading	Stored run (11 analyses)	Rerun
Size and sensitivity screens (Amendment 1)	S, DS, C1, C2, F1, F2, ES, EA	S, C1, C2, F1, F2, ES
Of these, not flagged in a prognostic cell	F1, ES, EA	F1, ES
As first specified (strawman S, symmetry screen on)	C2, F1	C1, F1, ES

Under Amendment 1 the strawman DS fails the size screen, so the strawman rule (Section 7, item 5) is not reached; DS was in any case flagged in the prognostic cells. Under the reading as first specified, S fails the symmetry screen on the

compound-symmetry curve in the rerun (it passed in the stored run), and the survivors change from C2 and F1 to C1, F1 and ES. The symmetry screen thus changes its verdict between two runs of the same design, as expected of a screen that tests Monte Carlo noise (Section 5; `docs/48`, Section 4).

Maximin shortfall

Table 12: Rerun under the refined process: maximin shortfall in size-adjusted power among the survivors of the size and sensitivity screens (eleven analyses; the best analysis in each cell is taken over all survivors, flagged ones included)

Analysis	Non-prognostic cells, rerun (stored)	Worst cell, rerun	All cells, rerun (stored)	Worst cell, rerun
C2 (flagged)	0.022 (0.040)	AR(1) curve, 0.45	0.058 (0.040)	TV constant
C1 (flagged)	0.032 (0.034)	measurement error	0.032 (0.064)	measurement error
F2 (flagged)	0.164 (0.160)	Weibull 2	0.208 (0.244)	TV growing
S (flagged)	0.172 (0.144)	Weibull 2	0.172 (0.220)	Weibull 2
DS (flagged)	fails size (0.182)		fails size (0.214)	
F1	0.240 (0.242)	Weibull 0.5	0.348 (0.358)	placebo
ES	0.266 (0.278)	Weibull 0.5	0.498 (0.544)	TV growing
EA	fails size (0.318)		fails size (0.528)	

Among the eleven analyses, F1 has the smallest maximin shortfall of the unflagged survivors in both readings: 0.240 against 0.266 for ES over the non-prognostic cells, a lead of 0.026, just outside the tie margin of 0.02, and 0.348 against 0.498 over all cells. F1's own shortfall is almost the same as in the stored run (0.240 against 0.242). Whether G1 or G2 again come within the tie margin of F1, as G2 did in the stored run (0.232 against 0.242), is taken up in the subsections "Fourteen-analysis decision" and "Fresh trials for the F1, G1, G2 choice" below.

Power

Table 13: Rerun under the refined process: size-adjusted power (unadjusted in parentheses) in the two reference alternative cells, $N = 70$, rerun and stored run

Analysis	AR(1), $c_{bm} = 0.30$, rerun	stored	Compound symmetry, $c_{bm} = 0.25$, rerun	stored
S	0.504 (0.506)	0.514 (0.508)	0.490 (0.498)	0.554 (0.554)
DS	0.502 (0.522)	0.486 (0.516)	0.372 (0.404)	0.480 (0.476)
C1	0.612 (0.594)	0.578 (0.602)	0.624 (0.632)	0.660 (0.658)
C2	0.606 (0.602)	0.606 (0.606)	0.616 (0.604)	0.656 (0.630)
F1	0.448 (0.424)	0.364 (0.410)	0.510 (0.534)	0.522 (0.544)
F2	0.482 (0.490)	0.516 (0.496)	0.488 (0.494)	0.552 (0.552)
ES	0.376 (0.392)	0.364 (0.412)	0.602 (0.572)	0.662 (0.634)
EA	0.390 (0.392)	0.306 (0.392)	0.564 (0.548)	0.602 (0.596)
A	0.556 (0.562)	0.534 (0.566)	0.566 (0.556)	0.618 (0.618)
DA	0.540 (0.568)	0.498 (0.592)	0.524 (0.542)	0.572 (0.596)
R	0.448 (0.482)	0.398 (0.476)	0.488 (0.546)	0.596 (0.594)

Table 14: Rerun under the refined process: size-adjusted power at $c_{bm} = 0.30$ in the stress cells for S, C1 and the two unflagged survivors F1 and ES (stored run in parentheses)

Cell	S	C1	F1	ES
$N = 150$	0.840 (0.828)	0.906 (0.918)	0.758 (0.748)	0.746 (0.704)
$N = 35$	0.258 (0.222)	0.360 (0.272)	0.208 (0.192)	0.188 (0.206)
$t_{1/2} = 0$	0.498 (0.594)	0.556 (0.666)	0.398 (0.496)	0.428 (0.504)
$t_{1/2} = 1$ week	0.374 (0.388)	0.454 (0.502)	0.274 (0.272)	0.276 (0.288)
Weibull decay, shape 0.5	0.392 (0.404)	0.522 (0.514)	0.288 (0.308)	0.262 (0.270)
Weibull decay, shape 2	0.494 (0.588)	0.666 (0.682)	0.498 (0.504)	0.466 (0.458)
Measurement error	0.396 (0.400)	0.462 (0.444)	0.282 (0.318)	0.286 (0.336)
20% MCAR dropout	0.446 (0.426)	0.556 (0.542)	0.356 (0.378)	0.352 (0.360)
20% MAR dropout	0.458 (0.418)	0.508 (0.518)	0.344 (0.382)	0.356 (0.356)
Prognostic, TV constant	0.598 (0.606)	0.752 (0.774)	0.456 (0.482)	0.402 (0.370)
Prognostic, TV growing	0.544 (0.516)	0.684 (0.672)	0.430 (0.378)	0.206 (0.192)

Cell	S	C1	F1	ES
Prognostic, placebo	0.586 (0.652)	0.746 (0.736)	0.398 (0.446)	0.448 (0.498)
Prognostic, baseline	0.490 (0.444)	0.582 (0.552)	0.416 (0.396)	0.406 (0.336)

F1 against S. At the AR(1) reference cell F1's size-adjusted power is 0.448 against 0.504 for S, 11% less, where the stored run gave 0.364 against 0.514, 29% less (Section 1, item 4). F1 rose by 0.084 ($z = 2.7$) while S changed by -0.010. Over the 20 non-prognostic alternative cells with $c_{bm} > 0$, the ratio of F1's to S's size-adjusted power averages 0.88 in the rerun and 0.83 in the stored run. The stored 29% was therefore larger than F1's typical deficit (17% on average in the stored run, 12% in the rerun), and part of it is plausibly Monte Carlo error in a single cell of 500 trials (inferred: the two runs are independent, and the rerun cannot separate the effect of the DGP change from chance). Under compound symmetry F1 is now slightly ahead of S (0.510 against 0.490; stored 0.522 against 0.554).

Paired comparisons within the rerun (McNemar test at 0.05 on paired size-adjusted rejections, 22 non-prognostic alternative cells):

- F1 against DS: F1 lower in 12 cells and higher in 3 (stored run: 15 and 0);
- F1 against S: F1 lower in 12 and higher in none (stored: 13 and 0);
- ES against S: ES lower in 14 and higher in 5 (stored: 15 and 3);
- C1 against S: C1 higher in 20 and lower in none (stored: 20 and 0).

Reading.

- **The cost of robustness is smaller than the stored run suggested at the AR(1) reference cell,** but it remains: F1 has significantly lower power than S in more than half of the non-prognostic cells, and the largest deficits are in the measurement-error, dropout, $t_{1/2} = 1$ and Weibull 0.5 cells (F1/S ratios 0.71 to 0.80).
- **Cell-level shifts between the runs are of the size expected from Monte Carlo error.** The mean change in unadjusted power over the 11 analyses ranges from -0.048 (compound-symmetry curve at 0.25) to $+0.054$ (prognostic baseline cell) over the 26 alternative cells, against an MCSE of about 0.03 for the difference of two independent cells of 500 trials. The changes come in blocks (for example every analysis lower at $t_{1/2} = 0$, a cell whose means anchoring does not change), because all analyses in a cell share the same trials. The $t_{1/2} = 0$ shift can only come from the allocation change or from chance; which of the two was not separated (a run with `--allocation fixed` would separate them).

- **Choosing the half-life by AIC spreads further.** The share of DS fits choosing $t^a_{1/2} = 0.5$ falls from 0.299 to 0.250, and the shares at 0 and 2 weeks rise (0.170 to 0.216, and 0.179 to 0.213).

Fourteen-analysis decision (jobs 21 and 22; added 2026-10-15)

G1 and G2 (simulation-server job 21, `--extra`) and F1A (job 22, `--extra2`) were rerun on the trials of job 15, so the fourteen analyses are paired (`replicates-full-extra-anchored.rds`, `replicates-full-extra2-anchored.rds`). The pairing was checked on the S arm, which every run refits: 27,997 of the 28,000 S estimates are identical between job 15 and job 21, and the other 3 are failed fits in both (verified). The decision was applied with `12-strawman-stress-decision.R --strawman DS --no-symmetry --with-extras --tag -anchored`, which writes `decision-*-ds-nosym-all14-anchored.csv` (committed `a4216f52`); the post hoc arm summaries are `posthoc-extra-anchored-*.csv` and `posthoc-extra2-anchored-*.csv` (`14-posthoc-arm-summary.R`).

Table 15: Rerun under the refined process, fourteen analyses: size, prognostic flag and maximin shortfall of the robust analyses (stored run in parentheses; the shortfall is measured against the best survivor in each cell, flagged survivors included)

Analysis	Pooled size, rerun (stored)	Largest cell, rerun	Prognostic flag	Maximin, non-prognostic, rerun (stored)	Maximin, all cells, rerun (stored)
G1	0.0503 (0.049)	0.062 (meas. error)	no	0.236 (0.252)	0.350 (0.308)
G2	0.0499 (0.047)	0.062 (Weibull 0.5)	no	0.214 (0.232)	0.342 (0.312)
F1A	0.0592 (0.057)	0.078 (Weibull 0.5)	no	fails size	fails size
F1	0.0514 (0.051)	0.064 ($t_{1/2} = 0$)	no	0.240 (0.242)	0.348 (0.358)
ES	0.0490 (0.048)	0.060 (Weibull 0.5)	no	0.266 (0.278)	0.498 (0.544)

Reading.

- **Survivors.** The size and sensitivity screens (Amendment 1) leave S, C1, C2, ES, F1, F2, G1 and G2. F1A fails size (pooled 0.059, outside the band 0.0458 to 0.0542, and 0.078 in the Weibull 0.5 cell against the cap of 0.075), as in the stored run; DS and EA fail as in the eleven-analysis rerun. Of the survivors, ES, F1, G1 and G2 are not flagged in any prognostic null cell (G1 and G2 at most 0.063).
- **Maximin shortfall.** Over the non-prognostic alternative cells the order is G2 0.214, G1 0.236, F1 0.240, ES 0.266. Under the rule as written, G2 leads F1 by 0.026, beyond the tie margin of 0.02, and would be selected; G1 ties with F1. Over all cells G2 (0.342), F1 (0.348) and G1 (0.350) lie within 0.008 of one another. In the stored run G2 led F1 by 0.010, within the margin.
- **Power at the AR(1) reference cell** ($c_{bm} = 0.30$, size adjusted): S 0.504, F1 0.448, G2 0.436, G1 0.426, F1A 0.404.

A lead of 0.026 is of the order of the Monte Carlo error of a maximin shortfall at 500 trials per alternative cell (inferred at this point: the MCSE of one size-adjusted power near 0.4 is about 0.022, and the maximin takes the worst of 22 such cells). Rather than select G2 on a margin of that size, fresh trials were run.

Fresh trials for the F1, G1, G2 choice (job 37; added 2026-10-15)

Job 37 (`11-strawman-stress-test.R --extra4`, driver committed `e24d71a6`, data `a3ea8f5f`) fitted the eight survivors S, C1, C2, ES, F1, F2, G1 and G2 to 1,500 fresh trials in each of the 41 cells (seed base 47261012, distinct from jobs 15, 21 and 22), under the refined process (`replicates-full-extra4-anchored.rds`, `summary-full-extra4-anchored.csv`). There were 104 failed fits in 492,000. The run is post hoc: it was made after the decision above had been seen, to settle a choice that lay within Monte Carlo error.

The unchanged decision script was applied to job 37 alone and to the trials of job 37 pooled with those of jobs 15 and 21 (job 37 replicate ids offset by 10,000; 2,500 trials per null cell and 2,000 per alternative cell). The shortfalls were bootstrapped by resampling trials within cell, paired across analyses (300 resamples). The scripts are `15-extra4-pooled-inputs.R`, `15-extra4-bootstrap.R` and `15-extra4-run.sh` in `analysis/scripts/quick-sim/carryover-closed-form/`; the decision files and logs are in `analysis/data/quick-sim/carryover-closed-form/strawman-stress/extr` (`decision-new.txt`, `decision-pooled.txt`, `boot.txt`, and the decision CSVs under `new/` and `pooled/`). A rerun of `15-extra4-run.sh` reproduced every output file of the first, scratch computation byte for byte (verified 2026-10-15).

Table 16: Job 37 and job 37 pooled with jobs 15 and 21: size over the 11 non-prognostic null cells, prognostic flag and maximin shortfall in size-adjusted power (the shortfall is measured against the best survivor in each cell; the bootstrap SE is reported for the three analyses in contention)

Analysis	Pooled size, job 37	Pooled size, all trials (largest cell)	Prognostic flag, all trials	Maximin, non-prognostic, job 37	Maximin, non-prognostic, all trials (bootstrap SE)	Maximin, all cells, all trials
F1	0.0487	0.0497 (0.057)	no	0.257	0.2155 (0.014)	0.373
G1	0.0482	0.0491 (0.056)	no	0.239	0.2205 (0.015)	0.3585
G2	0.0474	0.0484 (0.059)	no	0.245	0.2205 (0.017)	0.359
ES	0.0495	0.0493 (0.057)	no	0.243	0.235	0.4385
S	0.0496	0.0504 (0.059)	yes	0.151	0.122	0.133
F2	0.0504	0.0510 (0.059)	yes	0.157	0.1415	0.1455
C1	0.0487	0.0504 (0.060)	yes	0.000	0.0025	0.0025
C2	0.0458	0.0471 (0.051)	yes	fails size	fails size	fails size

Reading.

- **Size.** F1, G1 and G2 hold their size: over the 27,500 pooled null trials of the 11 non-prognostic null cells their rates are 0.0497, 0.0491 and 0.0484 (band 0.0474 to 0.0526), with no cell above 0.059, and they stay at or below 0.062 in every prognostic null cell (ES at most 0.063). C2 now fails the size screen as conservative (pooled 0.0471, below the band; 0.0458 in job 37 alone), so the pooled survivors are S, C1, ES, F1, F2, G1 and G2.
- **Maximin shortfall.** On the fresh trials alone the order is G1 0.239, ES 0.243, G2 0.245, F1 0.257: G1, G2 and F1 lie within 0.018. On all trials pooled it is F1 0.2155, G1 0.2205, G2 0.2205, ES 0.235, with F1 now first by 0.005. The three worst cells are all the measurement-error cell. Over all cells G1 (0.3585) and G2 (0.359) lead F1 (0.373) by 0.014, also within the margin.
- **Monte Carlo error.** The paired bootstrap SE of the F1 to G1, F1 to G2 and G1 to G2 differences in maximin shortfall is 0.012 to 0.016 on the

pooled trials (0.011 to 0.017 on job 37 alone). The tie margin of 0.02 is therefore not much larger than one SE of the difference at 2,000 trials per cell, and smaller than its SE at 500 trials per cell (about twice as large). This is why the first rerun appeared to favor G2: its lead of 0.026 does not recur on fresh trials (inferred from the two independent sets of trials and the bootstrap SE). (The bootstrap takes the best analysis in each cell over all eight, C2 included, so its ES shortfall, 0.2455 pooled, differs from the decision script's 0.235; the F1, G1 and G2 values agree.)

- **Power at the AR(1) reference cell** ($c_{bm} = 0.30$, all trials pooled, 2,000 per cell): S 0.496, G2 0.444, F1 0.431, G1 0.426; F1 has 13% less size-adjusted power than S (11% on the 500 trials of job 15 alone).

Decision (2026-10-15)

Under the rule of Section 7 on the pooled trials, F1, G1 and G2 lie within the tie margin of 0.02 and the tie-break (fewer covariance parameters) selects **F1**. F1 is retained as the recommended analysis for the biomarker-treatment interaction under the refined process. G1 and G2 are post hoc alternatives that tie with it; the lead of G2 on the 28,000 rerun trials is attributed to Monte Carlo error. Because job 37 was run after the first decision had been seen, and the margin is small against Monte Carlo error, the order of F1, G1 and G2 is not established by this study; an independent study with more trials per cell would be needed to separate them.

Not done: paired McNemar comparisons of F1 against G1 and G2 in each cell (the decision script runs them only against a surviving strawman); and the 1,500 fresh trials did not include F1A, DS, EA, A, DA or R, which had already failed a screen.

12. References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine* 2019;38(11):2074-2102. doi:10.1002/sim.8086.
2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2:13. doi:10.3389/fdgth.2020.00013.
3. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology* 2002;28(2):169-181.
4. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics* 2018;36(4):672-683. doi:10.1080/07350015.2016.1247004.

5. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics* 1997;53(3):983-997. doi:10.2307/2533558.
6. pmsimstats team. `docs/44`, `docs/45`, `docs/46`, `docs/48`, `docs/49`.